

Artificial Intelligence & Machine Learning

Comprehensive Master Reference Guide: Core Concepts & Crisp Definitions

1. Foundations & Supervised Learning

Concept	Crisp Definition
Linear Regression	A supervised learning algorithm used to predict a continuous target variable y by fitting a linear equation to observed data: $y = wx + b$.
Logistic Regression	A classification algorithm used to estimate the probability of a binary outcome using the sigmoid function: $P(y=1 x) = \sigma(wx + b)$.
Decision Trees	Non-parametric supervised models that split data into branches based on feature threshold rules to maximize information gain or minimize Gini impurity.
Random Forest	An ensemble learning method constructing multiple decision trees during training and outputting the mode (classification) or mean (regression) of individual trees.
Support Vector Machines (SVM)	Classifiers that find an optimal hyperplane maximizing the margin between data points of different classes, using kernel tricks for non-linear data.

2. Unsupervised Learning & Dimensionality Reduction

Concept	Crisp Definition
K-Means Clustering	An iterative unsupervised algorithm that partitions N observations into K clusters by minimizing within-cluster variance around centroids.
Principal Component Analysis (PCA)	A linear dimensionality reduction technique that transforms data into orthogonal axes (principal components) maximizing variance.
DBSCAN	Density-Based Spatial Clustering of Applications with Noise; groups high-density regions into clusters and flags low-density points as outliers.
Autoencoders	Neural networks trained to compress input data into a lower-dimensional bottleneck layer and reconstruct it, learning efficient latent representations.

3. Deep Learning & Neural Networks

Concept	Crisp Definition
Backpropagation	The core algorithm for training neural networks; computes the gradient of the loss function with respect to weights via the chain rule.

Convolutional Neural Networks (CNN)	Deep networks utilizing spatial convolution filters, ideal for grid-like data such as images and spatial matrices.
Recurrent Neural Networks (RNN)	Neural networks with cyclic connections designed for sequential data like time series or text, prone to vanishing gradients.
Long Short-Term Memory (LSTM)	An advanced RNN architecture utilizing input, forget, and output gates to mitigate vanishing gradients over long sequences.

4. Transformers & Generative AI

Concept	Crisp Definition
Self-Attention Mechanism	Computes dynamic relationships between all tokens in a sequence simultaneously using Query, Key, and Value projections: $Attention(Q,K,V) = softmax(QK^T / \sqrt{d_k})V$.
Retrieval-Augmented Generation (RAG)	An architecture combining external vector retrieval with LLM generation to ground responses in factual, enterprise knowledge bases.
LoRA (Low-Rank Adaptation)	A parameter-efficient fine-tuning technique that freezes base model weights and injects trainable rank decomposition matrices into layers.
Tokenization & BPE	The process of breaking raw text into subword units using Byte-Pair Encoding, balancing vocabulary size with unknown word handling.

5. Model Evaluation & Optimization

Concept	Crisp Definition
Bias-Variance Tradeoff	The balance between model simplicity (high bias, underfitting) and complexity (high variance, overfitting) impacting generalization.
Precision & Recall	Precision measures exactness ($TP / (TP + FP)$); Recall measures completeness ($TP / (TP + FN)$). F1-Score is their harmonic mean.
ROC-AUC	Receiver Operating Characteristic curve plotting True Positive Rate vs. False Positive Rate across thresholds; AUC measures overall ranking capability.
Regularization (L1/L2)	Techniques adding penalty terms to loss functions (L1/Lasso for sparsity, L2/Ridge for weight shrinkage) to prevent overfitting.